BIOT Module 4

Dhairya Bhatia

Disease and data choice, with biological interpretation

Disease: Alzheimer's disease (AD), carried over from Modules 2 and 3. Question: does each target's predicted 3D structure have a place where a drug could bind, and can I trust that structure?

Gene	UniProt	Module 2 rank	Module 3 prior
ITGAX (CD11c)	P20702	6	antibody, 0.59
APLNR (apelin receptor)	P35414	10	small molecule, 0.68

Why I chose them:

- **ITGAX:** found on the brain immune cells (microglia) that gather around plaques. A related protein, LFA-1, has a drug pocket in the same domain.
- **APLNR:** a GPCR, the classic pill-target family, and my strongest small-molecule prior. A crystal structure exists with a small drug bound.

Data used:

AlphaFold DB v6 predictions and PAE maps for both proteins, and experimental structures from the PDB: 4NEH chain A for ITGAX (2.75 Å, covers 1,084 residues) and 7SUS chain A for APLNR (2.7 Å, with the small-molecule agonist 8EH bound). I chose structures with the most coverage and, for APLNR, one with a real drug bound, so the pocket call could be checked against fact. All numbers come from my notebook.

What the results mean biologically:

- **APLNR:** the surprising hit from Module 2 turns out to be the most drug-ready. A real drug binds a confident pocket in the predicted structure, so a small molecule could act on the vascular signalling that APLNR controls.
- **ITGAX:** a known marker of the microglia that gather around plaques, but its structure offers no convincing drug pocket. Its route is an antibody, which matches its Module 3 prior.

1. Which parts can I trust?

Gene	Residues	Mean pLDDT	> 90	< 50	Low-confidence stretches	Domains found
ITGAX	1,163	88.0	65%	3.4%	1–18, 1145–1163	1 (20–1130)
APLNR	380	81.7	57%	15.5%	1–16, 334–343, 354–380	1 (19–325)

APLNR:

- *Trusted:* the seven helices that cross the membrane (residues 19–325) are confident and form one rigid block (mean PAE 5.8 Å), so I trust pockets there.
- *Not trusted:* the two tails. The front tail (1–16) is a loose end outside the cell. The back tail (after 334) sits inside the cell, where GPCRs are tagged to switch off, and it is floppy by design.
- *Two small dips:* the inside loops ICL2 (about 135–145) and ICL3 (about 230–240). The scientists who solved 7SUS replaced ICL3 with another protein (rubredoxin) because it was too floppy to crystallise, so the model and the experiment agree.

ITGAX:

- *Trusted:* almost everything except the signal peptide (1–18) and the tail inside the cell (1145–1163).

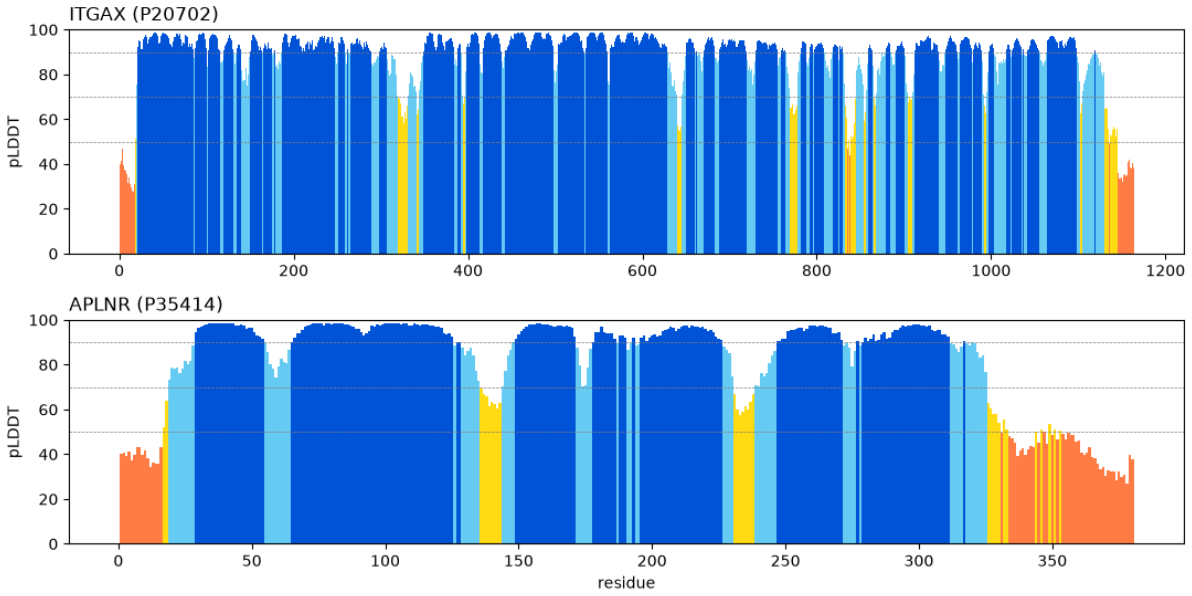


Figure 1: pLDDT along each protein. Dark blue means very confident, orange means very low confidence.

- *The domain finder is misleading:* it reports one domain, because it joins blocks through just a few confident links. The PAE map shows separate pieces.
  - *Within each piece:* confident. The β-propeller has a mean PAE of 4.7 Å and the I domain 4.8 Å.
  - *Between pieces:* uncertain. The I domain against the propeller has a mean PAE of 17.5 Å, and the head against the legs 20.5 Å.
- *What I trust for pockets:* by the slide 12 decision table, each piece has the right shape but their arrangement is uncertain. I trust pockets inside one piece, not pockets that span two.
- *This fits the biology:* integrins change shape when they switch on, so their pieces really do move.

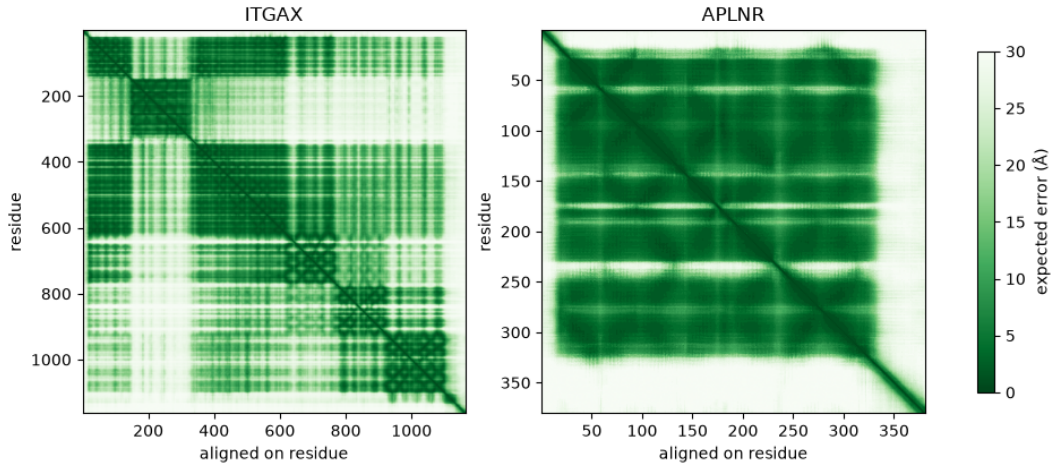


Figure 2: PAE maps. Dark squares are confident pieces. ITGAX has pale areas between its I domain (about 150–330), propeller and legs; APLNR is one dark block.

2. Does the prediction match experiment?

I paired residues by sequence alignment, and used only identical, confident pairs (pLDDT ≥ 70).

Target and structure	Region	Residues	RMSD
ITGAX vs 4NEH	Whole protein	1,020	0.86 Å
	Propeller / I domain / legs	265 / 178 / 577	0.25 / 0.32 / 0.64 Å
ITGAX vs 3K6S (older crystal)	Whole protein / propeller / I domain	—	8.53 / 1.77 / 1.82 Å
APLNR vs 7SUS	Helix bundle, insert removed	268	1.11 Å

ITGAX:

- *Against 4NEH:* the prediction matches almost exactly (0.86 Å). The biggest difference is a small loop in the legs (residues 907–912, up to 5.3 Å).
- *Against an older crystal, 3K6S:* each piece still matches within 2 Å, but the whole protein is 8.5 Å off.
- *What that means:* the two real structures disagree on how the pieces sit, which is exactly what the high PAE between the pieces warned about. AlphaFold picked one real arrangement, but the arrangement is not fixed.

APLNR:

- *First result:* 2.83 Å. The alignment had wrongly paired three APLNR residues (230, 239, 242) with the inserted rubredoxin protein, which happened to have matching letters. That is a pairing mistake, not a prediction error.
- *After removing those pairs:* 1.11 Å, and only 13 of 268 residues are more than 2 Å apart.
- *Where they differ:* the remaining differences are in loops: ICL2 (134–135), an outer loop (175–176), and residue 243 next to the insert. The crystal has things the prediction lacks: the inserted protein, seven stabilising mutations and a bound drug.

3. Pockets and druggability calls

Pockets were rejected if their lining had mean pLDDT below 70, and marked suspect if PAE among the lining residues was above 10 Å.

Gene	Pocket	Drug score	Volume (Å³)	Mean pLDDT	PAE within	Verdict
ITGAX	4	0.489	284	89.0	2.6	keep
	79	0.470	927	89.3	3.2	keep
	3	0.098	517	92.2	1.8	keep
	26	0.093	434	90.3	2.6	keep
	71	0.083	371	92.7	1.9	keep
APLNR	9	0.304	428	94.2	2.0	keep
	22	0.182	364	91.2	2.2	keep
	10	0.111	951	82.0	4.3	keep
	2	0.015	1,761	94.0	3.1	keep
	1	0.010	714	78.4	5.5	keep

APLNR, "Structure supports a small-molecule pocket":

- *Confident site:* the residues touching the real drug have mean pLDDT 95.1.
- *The pocket is already there:* I moved the drug from the crystal into the prediction. Only 6 of its 37 atoms bump into the model, all against two side chains (W85 and Y299), which just need to shift slightly when the drug binds.
- *Proof:* a real small-molecule drug binds this site in the crystal.
- *Catch:* fpocket gives this pocket a low score; Section 6 explains why.

ITGAX, "Possible but uncertain", with an antibody as the better route:

- *Best pockets:* the two top-scoring pockets (4 and 79, scores 0.49 and 0.47) are kept and sit on the propeller. They are not covered by the partner protein or by sugars in 4NEH (both 11–14 Å away).
- *Why not a stronger call:* no drug is known for either pocket, and both scores are below 0.5. Pocket 79 also stretches across two pieces (residues 104–388), which PAE says I can't trust.

- *The pocket Module 3 hoped for is missing:* in LFA-1, the drug lovastatin binds a pocket in the I domain. I copied lovastatin from the LFA-1 crystal (1CQP) into ITGAX.
  - *Control:* with the drug removed, fpocket finds the pocket in LFA-1 (score 0.44, holding 72% of the drug), so the method works.
  - *ITGAX:* lovastatin bumps into the protein with 22 of its 29 atoms in the prediction, and 27 of 29 in the real ITGAX crystal (1N3Y). No pocket holds it.
  - *Conclusion:* prediction and experiment agree that ITGAX does not have LFA-1's pocket.
- *Natural binding site:* the metal-binding site (MIDAS) is shallow and polar, with no pocket.
- *Better route, an antibody:* ITGAX is a large protein on the outside of the cell with flat surfaces, which suits antibodies. For AD, the antibody would also need a way across the blood–brain barrier, because microglia are inside the brain.

4. How this fits with Modules 2 and 3

Both calls match the Module 3 prior. APLNR looked like a pill target (0.68), and the structure confirms a real, confident pocket. ITGAX looked like an antibody target (0.59), and the structure rules out the one pill idea I had for it. I would keep both, with different jobs:

- *APLNR, the small-molecule candidate:* its Module 2 rank (10) rests on a strong RNA change (log2FC 1.28) but weak protein support ($p = 0.068$), so I would firm up its AD link before starting chemistry.
- *ITGAX, the antibody candidate:* it keeps its higher disease ranking (6, with RNA and protein agreeing).

5. Discordance and limitations

Discordant results:

- *ITGAX, prediction versus an older crystal:* the prediction matches 4NEH almost exactly (0.86 Å) but is 8.5 Å off 3K6S, even though each piece matches within 2 Å. The two real structures disagree on how the pieces sit, which is what the high PAE between pieces warned about.
- *APLNR, pocket score versus the real drug site:* fpocket's top pocket (9) misses the drug entirely, while the pocket that holds the real drug ranks only 4th (score 0.015). Section 6 explains why.
- *RNA and protein (from Module 2):* both targets move the same way in RNA and protein, so neither is discordant there. APLNR's protein support is weak, though ($p = 0.068$).

What this lets me claim: APLNR has a confident pocket where a real drug binds, and ITGAX lacks the pocket that LFA-1 has. *What it does not:* that either protein is a good AD drug target, that a drug would reach it in the brain, or that the pockets exist in every shape these proteins take.

Limits (slide 22) that apply to my targets:

- *One snapshot:*
  - *The limit:* each prediction is one frozen shape. APLNR is shown in its "off" state; in a side check outside the notebook, an "on" structure with a G protein (7W0L) differed most at the inner end of helix 6 (residues 235–240). ITGAX's two crystals disagree by 8.5 Å.
  - *What would help:* simulations (molecular dynamics) of APLNR with and without the drug, and docking into the "on" structures. For ITGAX, compare bent and extended structures before trusting any pocket that spans two pieces.
- *Missing pieces and no membrane:*
  - *The limit:* both models are single chains. ITGAX has no partner protein (β2), no metal ion and no sugars. APLNR has no membrane, even though the crystal shows a lipid sitting against the same pocket.
  - *What would help:* model ITGAX with its partner and metal (AlphaFold-Multimer or Boltz-2 on the cluster), and simulate APLNR inside a membrane.
- *Hidden pockets:*
  - *The limit:* ITGAX's missing pocket might open only in another shape.
  - *What would help:* a fragment screen or a thermal-shift test on the purified I domain.

6. Ligand site (APLNR, 7SUS)

Is my best kept pocket where the drug sits? No. 7SUS contains one drug molecule (8EH) and one lipid (OLC) left over from crystallisation. The drug is the one that matters: it sits deep in the helix bundle, touching 19 residues including

W85, Y271 and Y299.

- *Best kept pocket:* pocket 9 (rank 1, score 0.304) covers 0% of the drug site. It is lower down in the bundle, about 24 Å away, and no drug is known to bind there.
- *The real drug site:* pocket 2 covers 68% of the drug, but it ranks only 4th, with a score of 0.015, even though its confidence is high (pLDDT 94.0).

Why the real site scores low: APLNR's natural partner is a peptide that enters from outside the cell, so the pocket is wide open on that side. fpocket merged the deep drug site and this open entrance into one huge cavity (1,761 Å³), and its score punishes big, open pockets. In the prediction, W85 and Y299 also partly fill the space. So it is the confidence checks and the real crystal, not fpocket's score, that support the APLNR call.

Anchor check: p53 (lab Part 2)

- *Match to experiment:* the p53 prediction matches 1TUP chain B at 0.68 Å over 192 confident residues.
- *Zinc ligands:* C176, H179, C238 and C242 hold the zinc.
- *DNA contacts:* R248 and R273 touch DNA directly, and R249 is near it.
- *Structural residues:* R175, Y220, G245 and R282 touch neither.
- *What this means for my Module 3 scores:* the DNA-contact mutation R248Q scored as more unusual (−9.55) than the structural mutation R175H (−5.97). The language model measures how unusual a change is in evolution, not what it breaks; only the structure shows that R248Q breaks a DNA contact while R175H loosens the fold near the zinc.